

betaMC: Internal Tests

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Tests

```
#> test-betaMC-beta-mc-est-mi
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0734 5 0.4295 0.4301 0.4329 0.5945 0.5956 0.5959
#> PCTGRT  0.3915 0.0677 5 0.2056 0.2063 0.2092 0.3539 0.3542 0.3543
#> PCTSUPP 0.2632 0.0807 5 0.1635 0.1661 0.1775 0.3601 0.3617 0.3621
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0734 5 0.4295 0.4301 0.4329 0.5945 0.5956 0.5959
#> PCTGRT  0.3915 0.0677 5 0.2056 0.2063 0.2092 0.3539 0.3542 0.3543
#> PCTSUPP 0.2632 0.0807 5 0.1635 0.1661 0.1775 0.3601 0.3617 0.3621
#> Test passed with 1 success.
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
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#> Test passed with 1 success.

#> test-betaMC-beta-mc-est

#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0701 5 0.3652 0.3657 0.3678 0.5235 0.5250 0.5254
#> PCTGRT  0.3915 0.0558 5 0.3502 0.3517 0.3585 0.4947 0.4976 0.4982
#> PCTSUPP 0.2632 0.0931 5 0.1234 0.1247 0.1305 0.3566 0.3630 0.3645
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0701 5 0.3652 0.3657 0.3678 0.5235 0.5250 0.5254
#> PCTGRT  0.3915 0.0558 5 0.3502 0.3517 0.3585 0.4947 0.4976 0.4982
#> PCTSUPP 0.2632 0.0931 5 0.1234 0.1247 0.1305 0.3566 0.3630 0.3645
#> Test passed with 1 success.
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
#> Test passed with 1 success.

#> test-betaMC-delta-r-sq-mc-est-mi

#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%

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#> NARTIC 0.1859 0.0750 5 0.0805 0.0812 0.0845 0.2688 0.2771 0.2789
#> PCTGRT 0.1177 0.0302 5 0.0244 0.0248 0.0267 0.0964 0.0970 0.0972
#> PCTSUPP 0.0569 0.0361 5 0.0186 0.0193 0.0221 0.1141 0.1176 0.1183
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.1859 0.0750 5 0.0805 0.0812 0.0845 0.2688 0.2771 0.2789
#> PCTGRT 0.1177 0.0302 5 0.0244 0.0248 0.0267 0.0964 0.0970 0.0972
#> PCTSUPP 0.0569 0.0361 5 0.0186 0.0193 0.0221 0.1141 0.1176 0.1183
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-delta-r-sq-mc-est

#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.1859 0.0584 5 0.0450 0.0460 0.0505 0.1960 0.2004 0.2014
#> PCTGRT 0.1177 0.0355 5 0.0870 0.0879 0.0916 0.1754 0.1758 0.1759
#> PCTSUPP 0.0569 0.0440 5 0.0106 0.0107 0.0113 0.1125 0.1185 0.1199
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.1859 0.0584 5 0.0450 0.0460 0.0505 0.1960 0.2004 0.2014
#> PCTGRT 0.1177 0.0355 5 0.0870 0.0879 0.0916 0.1754 0.1758 0.1759
#> PCTSUPP 0.0569 0.0440 5 0.0106 0.0107 0.0113 0.1125 0.1185 0.1199
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-diff-beta-mc-est-mi

#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC-PCTGRT 0.1037 0.1170 5 0.0798 0.0803 0.0827 0.3467 0.3523 0.3536

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#> NARTIC-PCTSUPP 0.2319 0.1419 5 0.0874 0.0876 0.0888 0.4057 0.4160 0.4183
#> PCTGRT-PCTSUPP 0.1282 0.1110 5 -0.1362 -0.1349 -0.1290 0.1481 0.1606 0.1634
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>          est      se R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT 0.1037 0.1170 5 0.0798 0.0803 0.0827 0.3467 0.3523 0.3536
#> NARTIC-PCTSUPP 0.2319 0.1419 5 0.0874 0.0876 0.0888 0.4057 0.4160 0.4183
#> PCTGRT-PCTSUPP 0.1282 0.1110 5 -0.1362 -0.1349 -0.1290 0.1481 0.1606 0.1634
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-diff-beta-mc-est

#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>          est      se R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT 0.1037 0.1073 5 -0.1330 -0.1319 -0.1269 0.1136 0.1157 0.1161
#> NARTIC-PCTSUPP 0.2319 0.1332 5 0.1016 0.1017 0.1021 0.3930 0.4003 0.4020
#> PCTGRT-PCTSUPP 0.1282 0.1350 5 -0.0142 -0.0107 0.0046 0.3101 0.3110 0.3112
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>          est      se R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT 0.1037 0.1073 5 -0.1330 -0.1319 -0.1269 0.1136 0.1157 0.1161
#> NARTIC-PCTSUPP 0.2319 0.1332 5 0.1016 0.1017 0.1021 0.3930 0.4003 0.4020
#> PCTGRT-PCTSUPP 0.1282 0.1350 5 -0.0142 -0.0107 0.0046 0.3101 0.3110 0.3112
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-mc-fixed-x-mi

#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.

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#> Test passed with 1 success.
#> Test passed with 1 success.
#> test-betaMC-mc-fixed-x
#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Call:
#> MC(object = object, R = 5L, decomposition = "chol", fixed_x = TRUE)
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4758462 0.4987145
#>
#> $sigmasq
#> [1] 0.5337949
#>
#> $vechsigmacapx
#> [1] 1.000000e+00 5.848434e-17 1.000000e+00
#>
#> $sigmacapx
#>      [,1]      [,2]
#> [1,] 1.000000e+00 5.848434e-17
#> [2,] 5.848434e-17 1.000000e+00
#>
#> $sigmaysq
#> [1] 1.008941
#>
#> $sigmayx
#> [1] 0.4758462 0.4987145
#>
#> $sigmacap
#>      [,1]      [,2]      [,3]
#> [1,] 1.0089407 4.758462e-01 4.987145e-01
#> [2,] 0.4758462 1.000000e+00 5.848434e-17
#> [3,] 0.4987145 5.848434e-17 1.000000e+00
#>
#> $pd
#> [1] TRUE

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#>
#> Call:
#> MC(object = object, R = 5L, decomposition = "svd", fixed_x = TRUE)
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4659576 0.5142679
#>
#> $sigmasq
#> [1] 0.5359596
#>
#> $vechsigmacapx
#> [1] 1.000000e+00 5.848434e-17 1.000000e+00
#>
#> $sigmacapx
#>           [,1]      [,2]
#> [1,] 1.000000e+00 5.848434e-17
#> [2,] 5.848434e-17 1.000000e+00
#>
#> $sigmaysq
#> [1] 1.017548
#>
#> $sigmayx
#> [1] 0.4659576 0.5142679
#>
#> $sigmacap
#>           [,1]      [,2]      [,3]
#> [1,] 1.0175476 4.659576e-01 5.142679e-01
#> [2,] 0.4659576 1.000000e+00 5.848434e-17
#> [3,] 0.5142679 5.848434e-17 1.000000e+00
#>
#> $pd
#> [1] TRUE
#>
#> Test passed with 1 success.
#> test-betaMC-mc-mi
#> MCMC(object = object, mi = mi, R = R, type = "mvn")
#> $mean
#>      b1      b2  sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> 0.4826 0.4891 0.5299 0.9983 0.0104 1.0038
#>
#> $var
#>      b1      b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> b1 5e-04 0e+00 0e+00 1e-04 0e+00 0.0000

```

```

#> b2          0e+00 7e-04 1e-04 1e-04 1e-04 -0.0002
#> sigmasq      0e+00 1e-04 4e-04 -1e-04 0e+00 0.0000
#> sigma1x1     1e-04 1e-04 -1e-04 2e-03 1e-04 0.0003
#> sigma2x1     0e+00 1e-04 0e+00 1e-04 1e-03 0.0002
#> sigma2x2     0e+00 -2e-04 0e+00 3e-04 2e-04 0.0018
#>
#> $bias
#>      b1      b2 sigmasq sigma1x1 sigma2x1 sigma2x2
#> -0.0020 0.0019 -0.0008 -0.0017 0.0004 0.0038
#>
#> $rmse
#>      b1      b2 sigmasq sigma1x1 sigma2x1 sigma2x2
#> 0.0220 0.0257 0.0207 0.0441 0.0322 0.0428
#>
#> $location
#>      b1      b2 sigmasq sigma1x1 sigma2x1 sigma2x2
#> 0.4845 0.4872 0.5307 1.0000 0.0100 1.0000
#>
#> $scale
#>      b1      b2 sigmasq sigma1x1 sigma2x1 sigma2x2
#> b1      5e-04 0e+00 0e+00 0.000 0.000 0.000
#> b2      0e+00 5e-04 0e+00 0.000 0.000 0.000
#> sigmasq 0e+00 0e+00 6e-04 0.000 0.000 0.000
#> sigma1x1 0e+00 0e+00 0e+00 0.002 0.000 0.000
#> sigma2x1 0e+00 0e+00 0e+00 0.000 0.001 0.000
#> sigma2x2 0e+00 0e+00 0e+00 0.000 0.000 0.002
#>
#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#>
#> test-betaMC-mc
#> MC(object = object, R = R, type = "mvn")
#> $mean
#>      b1      b2 sigmasq sigma1x1 sigma2x1 sigma2x2
#> 0.4838 0.4895 0.5296 0.9947 0.0074 1.0069
#>
#> $var
#>      b1      b2 sigmasq sigma1x1 sigma2x1 sigma2x2
#> b1      6e-04 1e-04 1e-04 0.0000 0.0001 0.0000

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#> b2          1e-04  5e-04 -1e-04 -0.0002  0.0000  0.0001
#> sigmasq      1e-04 -1e-04  6e-04 -0.0001 -0.0001 -0.0001
#> sigmax1x1    0e+00 -2e-04 -1e-04  0.0018  0.0003 -0.0002
#> sigmax2x1    1e-04  0e+00 -1e-04  0.0003  0.0012  0.0002
#> sigmax2x2    0e+00  1e-04 -1e-04 -0.0002  0.0002  0.0018
#>
#> $bias
#>      b1      b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> -0.0008  0.0023 -0.0010 -0.0053 -0.0025  0.0069
#>
#> $rmse
#>      b1      b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>  0.0236  0.0224  0.0248  0.0422  0.0348  0.0431
#>
#> $location
#>      b1      b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>  0.4845  0.4872  0.5307  1.0000  0.0100  1.0000
#>
#> $scale
#>      b1      b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> b1      5e-04  0e+00  0e+00  0.000  0.000  0.000
#> b2      0e+00  5e-04  0e+00  0.000  0.000  0.000
#> sigmasq  0e+00  0e+00  6e-04  0.000  0.000  0.000
#> sigmax1x1 0e+00  0e+00  0e+00  0.002  0.000  0.000
#> sigmax2x1 0e+00  0e+00  0e+00  0.000  0.001  0.000
#> sigmax2x2 0e+00  0e+00  0e+00  0.000  0.000  0.002
#>
#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Call:
#> MC(object = object, R = 5L, decomposition = "chol")
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4688013 0.4660633
#>
#> $sigmasq
#> [1] 0.5331191

```



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#>
#> $vechsigmacapx
#> [1] 0.9497147989 -0.0003547646 1.0075414443
#>
#> $sigmacapx
#>           [,1]           [,2]
#> [1,] 0.9497147989 -0.0003547646
#> [2,] -0.0003547646 1.0075414443
#>
#> $sigmayx
#> [1] 0.9605405
#>
#> $sigmayx
#> [1] 0.4450622 0.4694118
#>
#> $sigmacap
#>           [,1]           [,2]           [,3]
#> [1,] 0.9605405 0.4450622080 0.4694118229
#> [2,] 0.4450622 0.9497147989 -0.0003547646
#> [3,] 0.4694118 -0.0003547646 1.0075414443
#>
#> $pd
#> [1] TRUE
#>
#> Call:
#> MC(object = object, R = 5L, decomposition = "svd")
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4580301 0.4850555
#>
#> $sigmasq
#> [1] 0.5497989
#>
#> $vechsigmacapx
#> [1] 1.038296916 0.009804769 1.068440723
#>
#> $sigmacapx
#>           [,1]           [,2]
#> [1,] 1.038296916 0.009804769
#> [2,] 0.009804769 1.068440723
#>
#> $sigmayx
#> [1] 1.023363
#>

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#> $sigmayx
#> [1] 0.480327 0.522744
#>
#> $sigmacap
#>      [,1]      [,2]      [,3]
#> [1,] 1.023363 0.480327047 0.522743966
#> [2,] 0.480327 1.038296916 0.009804769
#> [3,] 0.522744 0.009804769 1.068440723
#>
#> $pd
#> [1] TRUE
#>
#> Test passed with 1 success.

#> test-betaMC-p-cor-mc-est-mi

#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5%  99.95%
#> NARTIC  0.4874 0.1085 5 0.3270 0.3271 0.3274 0.5766 0.5863 0.5885
#> PCTGRT  0.3757 0.0746 5 0.1301 0.1314 0.1369 0.3121 0.3149 0.3155
#> PCTSUPP 0.2254 0.1079 5 0.0849 0.0871 0.0966 0.3698 0.3780 0.3799
#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5%  99.95%
#> NARTIC  0.4874 0.1085 5 0.3270 0.3271 0.3274 0.5766 0.5863 0.5885
#> PCTGRT  0.3757 0.0746 5 0.1301 0.1314 0.1369 0.3121 0.3149 0.3155
#> PCTSUPP 0.2254 0.1079 5 0.0849 0.0871 0.0966 0.3698 0.3780 0.3799
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-p-cor-mc-est

#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5%  99.95%
#> NARTIC  0.4874 0.1162 5 0.2344 0.2354 0.2401 0.5105 0.5169 0.5184
#> PCTGRT  0.3757 0.0736 5 0.3104 0.3122 0.3203 0.4926 0.4933 0.4934

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```

#> PCTSUPP 0.2254 0.1260 5 0.0543 0.0553 0.0598 0.3601 0.3785 0.3826
#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.4874 0.1162 5 0.2344 0.2354 0.2401 0.5105 0.5169 0.5184
#> PCTGRT 0.3757 0.0736 5 0.3104 0.3122 0.3203 0.4926 0.4933 0.4934
#> PCTSUPP 0.2254 0.1260 5 0.0543 0.0553 0.0598 0.3601 0.3785 0.3826
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-r-sq-mc-est-mi

#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0312 5 0.7508 0.7517 0.7556 0.8343 0.8367 0.8373
#> adj 0.7906 0.0334 5 0.7330 0.7340 0.7382 0.8224 0.8251 0.8256
#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0312 5 0.7508 0.7517 0.7556 0.8343 0.8367 0.8373
#> adj 0.7906 0.0334 5 0.7330 0.7340 0.7382 0.8224 0.8251 0.8256
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-r-sq-mc-est

#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0346 5 0.7565 0.7574 0.7614 0.8496 0.8526 0.8533
#> adj 0.7906 0.0371 5 0.7391 0.7401 0.7444 0.8388 0.8421 0.8428
#> Call:
#> RSqMC(object = mc)
#>

```

```

#> R-squared and adjusted R-squared
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0346 5 0.7565 0.7574 0.7614 0.8496 0.8526 0.8533
#> adj 0.7906 0.0371 5 0.7391 0.7401 0.7444 0.8388 0.8421 0.8428
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-s-cor-mc-est-mi

#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.4312 0.0911 5 0.2837 0.2849 0.2900 0.5175 0.5261 0.5281
#> PCTGRT 0.3430 0.0642 5 0.1560 0.1572 0.1624 0.3105 0.3115 0.3118
#> PCTSUPP 0.2385 0.0752 5 0.1364 0.1381 0.1458 0.3371 0.3427 0.3440
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.4312 0.0911 5 0.2837 0.2849 0.2900 0.5175 0.5261 0.5281
#> PCTGRT 0.3430 0.0642 5 0.1560 0.1572 0.1624 0.3105 0.3115 0.3118
#> PCTSUPP 0.2385 0.0752 5 0.1364 0.1381 0.1458 0.3371 0.3427 0.3440
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-s-cor-mc-est

#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.4312 0.0892 5 0.2120 0.2139 0.2224 0.4423 0.4476 0.4487
#> PCTGRT 0.3430 0.0497 5 0.2950 0.2963 0.3019 0.4187 0.4193 0.4194
#> PCTSUPP 0.2385 0.0944 5 0.1029 0.1035 0.1061 0.3329 0.3438 0.3462
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"

```

```

#>          est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0892 5 0.2120 0.2139 0.2224 0.4423 0.4476 0.4487
#> PCTGRT  0.3430 0.0497 5 0.2950 0.2963 0.3019 0.4187 0.4193 0.4194
#> PCTSUPP 0.2385 0.0944 5 0.1029 0.1035 0.1061 0.3329 0.3438 0.3462
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-zzz-coverage

#>          beta1      beta2      beta3 sigmasq  sigmax1x1  sigmax2x1  sigmax3x1
#> sigmaysq  909.1981 257.2976 276.0367      1 0.007091036 0.03637752 0.01896371
#> sigmayx1 3507.1691 471.2058 510.5430      0 0.084208291 0.21599726 0.11260003
#> sigmayx2  471.2058 333.2295 150.9121      0 0.000000000 0.08420829 0.00000000
#> sigmayx3  510.5430 150.9121 554.4386      0 0.000000000 0.00000000 0.08420829
#> sigmax1x1  0.0000  0.0000  0.0000      0 1.000000000 0.00000000 0.00000000
#> sigmax2x1  0.0000  0.0000  0.0000      0 0.000000000 1.00000000 0.00000000
#> sigmax3x1  0.0000  0.0000  0.0000      0 0.000000000 0.00000000 1.00000000
#> sigmax2x2  0.0000  0.0000  0.0000      0 0.000000000 0.00000000 0.00000000
#> sigmax3x2  0.0000  0.0000  0.0000      0 0.000000000 0.00000000 0.00000000
#> sigmax3x3  0.0000  0.0000  0.0000      0 0.000000000 0.00000000 0.00000000
#>          sigmax2x2 sigmax3x2  sigmax3x3
#> sigmaysq  0.04665482 0.0486426 0.01267877
#> sigmayx1  0.00000000 0.0000000 0.00000000
#> sigmayx2  0.21599726 0.1126000 0.00000000
#> sigmayx3  0.00000000 0.2159973 0.11260003
#> sigmax1x1 0.00000000 0.0000000 0.00000000
#> sigmax2x1 0.00000000 0.0000000 0.00000000
#> sigmax3x1 0.00000000 0.0000000 0.00000000
#> sigmax2x2 1.00000000 0.0000000 0.00000000
#> sigmax3x2 0.00000000 1.0000000 0.00000000
#> sigmax3x3 0.00000000 0.0000000 1.00000000
#>          beta1      beta2      beta3 sigmasq
#> sigmaysq  909.1981 257.2976 276.0367      1
#> sigmayx1 3507.1691 471.2058 510.5430      0
#> sigmayx2  471.2058 333.2295 150.9121      0
#> sigmayx3  510.5430 150.9121 554.4386      0
#> sigmax1x1  0.0000  0.0000  0.0000      0
#> sigmax2x1  0.0000  0.0000  0.0000      0
#> sigmax3x1  0.0000  0.0000  0.0000      0
#> sigmax2x2  0.0000  0.0000  0.0000      0
#> sigmax3x2  0.0000  0.0000  0.0000      0
#> sigmax3x3  0.0000  0.0000  0.0000      0
#>          beta1      beta2      beta3      rsq  sigmax1x1  sigmax2x1
#> sigmaysq  909.1981 257.2976 276.0367 -126.0843 0.007091036 0.03637752
#> sigmayx1 3507.1691 471.2058 510.5430  0.0000 0.084208291 0.21599726
#> sigmayx2  471.2058 333.2295 150.9121  0.0000 0.000000000 0.08420829
#> sigmayx3  510.5430 150.9121 554.4386  0.0000 0.000000000 0.00000000

```

```

#> sigma1x1 0.0000 0.0000 0.0000 0.0000 1.000000000 0.00000000
#> sigma2x1 0.0000 0.0000 0.0000 0.0000 0.000000000 1.00000000
#> sigma3x1 0.0000 0.0000 0.0000 0.0000 0.000000000 0.00000000
#> sigma2x2 0.0000 0.0000 0.0000 0.0000 0.000000000 0.00000000
#> sigma3x2 0.0000 0.0000 0.0000 0.0000 0.000000000 0.00000000
#> sigma3x3 0.0000 0.0000 0.0000 0.0000 0.000000000 0.00000000
#>          sigma3x1 sigma2x2 sigma3x2 sigma3x3
#> sigma1sq 0.01896371 0.04665482 0.0486426 0.01267877
#> sigma1x1 0.11260003 0.00000000 0.0000000 0.00000000
#> sigma1x2 0.00000000 0.21599726 0.1126000 0.00000000
#> sigma1x3 0.08420829 0.00000000 0.2159973 0.11260003
#> sigma2x1 0.00000000 0.00000000 0.0000000 0.00000000
#> sigma2x2 0.00000000 0.00000000 0.0000000 0.00000000
#> sigma2x3 1.00000000 0.00000000 0.0000000 0.00000000
#> sigma3x2 0.00000000 1.00000000 0.0000000 0.00000000
#> sigma3x3 0.00000000 0.00000000 1.0000000 0.00000000
#>          beta1 beta2 beta3 rsq
#> sigma1sq 909.1981 257.2976 276.0367 -126.0843
#> sigma1x1 3507.1691 471.2058 510.5430 0.0000
#> sigma1x2 471.2058 333.2295 150.9121 0.0000
#> sigma1x3 510.5430 150.9121 554.4386 0.0000
#> sigma2x1 0.0000 0.0000 0.0000 0.0000
#> sigma2x2 0.0000 0.0000 0.0000 0.0000
#> sigma2x3 0.0000 0.0000 0.0000 0.0000
#> sigma3x2 0.0000 0.0000 0.0000 0.0000
#> sigma3x3 0.0000 0.0000 0.0000 0.0000
#> Test passed with 1 success.
#> [[1]]
#> [[1]] [[1]]
#> [[1]] [[1]]$value
#> [[1]] [[1]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]] [[1]]$visible
#> [1] TRUE
#>
#>
#> [[1]] [[2]]
#> [[1]] [[2]]$value
#> [[1]] [[2]]$value[[1]]
#> [1] TRUE
#>
#>
#>

```

```

#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[5]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[6]]
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[7]]

```

```

#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[8]]
#> [[1]][[8]]$value
#> [[1]][[8]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[8]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[9]]
#> [[1]][[9]]$value
#> [[1]][[9]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[9]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[10]]
#> [[1]][[10]]$value
#> [[1]][[10]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[10]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[11]]
#> [[1]][[11]]$value
#> [[1]][[11]]$value[[1]]
#> [1] TRUE
#>
#>
#>

```



```
#> [[1]][[11]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[12]]
#> [[1]][[12]]$value
#> [[1]][[12]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[12]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[13]]
#> [[1]][[13]]$value
#> [[1]][[13]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[13]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[14]]
#> [[1]][[14]]$value
#> [[1]][[14]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[14]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[15]]
#> [[1]][[15]]$value
#> [[1]][[15]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[15]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[16]]
```

```
#> [[1]][[16]]$value
#> [[1]][[16]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[16]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[17]]
#> [[1]][[17]]$value
#> [[1]][[17]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[17]]$visible
#> [1] TRUE
```

Environment

```
ls()
```

```
#> [1] "nas1982" "root"
```

Class

```
#> [[1]]  
#> [1] "data.frame"  
#>  
#> [[2]]  
#> [1] "root_criterion"
```

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>